

Lecture on Partial Fraction Decomposition

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October 2, 2024

1 Introduction to Partial Fractions

Partial fraction decomposition is a technique used to express a rational fraction as a sum of simpler fractions. This is particularly useful in integration and algebraic manipulation.

2 Types of Partial Fractions

Partial fractions can be classified based on the factors of the denominator.

2.1 Case 1: Linear Factors

Example: $\frac{P(x)}{(x-a)(x-b)}$

2.2 Case 2: Irreducible Quadratic Factors

Example: $\frac{P(x)}{(x^2+ax+b)}$

2.3 Case 3: Repeated Linear Factors

Example: $\frac{P(x)}{(x-a)^n}$

2.4 Case 4: Mixed Factors

Example: $\frac{P(x)}{(x-a)(x^2+bx+c)}$

3 Examples of Partial Fraction Decomposition

3.1 Case 1: Linear Factors

1. $\frac{3x+5}{(x-1)(x+2)}$
2. $\frac{2x+3}{(x-3)(x+4)}$

$$3. \frac{x^2+4}{(x-2)(x+1)}$$

$$4. \frac{5}{(x+3)(x-5)}$$

$$5. \frac{4x+7}{(x-1)(x+3)}$$

3.2 Case 2: Irreducible Quadratic Factors

$$1. \frac{x+2}{(x^2+1)(x-1)}$$

$$2. \frac{3x}{(x^2+2)(x+1)}$$

$$3. \frac{2x+5}{(x^2+4)(x-2)}$$

$$4. \frac{1}{(x^2+3)(x+1)}$$

$$5. \frac{4x+6}{(x^2+1)(x-3)}$$

3.3 Case 3: Repeated Linear Factors

$$1. \frac{5}{(x-2)^2}$$

$$2. \frac{3x+1}{(x+1)^3}$$

$$3. \frac{2}{(x-1)^3}$$

$$4. \frac{4}{(x+2)^2(x-1)}$$

$$5. \frac{6x}{(x-3)^3}$$

3.4 Case 4: Mixed Factors

$$1. \frac{x+3}{(x-1)(x^2+1)}$$

$$2. \frac{2x+1}{(x+2)(x^2+4)}$$

$$3. \frac{5}{(x+1)(x^2+2x+5)}$$

$$4. \frac{3x}{(x-2)(x^2+1)}$$

$$5. \frac{1}{(x-3)(x^2+2)}$$

4 1. Linear Factors

Example:

$$\frac{2x + 5}{(x - 1)(x + 3)}$$

Solution:

$$\frac{A}{x - 1} + \frac{B}{x + 3}$$

$$\begin{aligned} 2x + 5 &= A(x + 3) + B(x - 1) \\ &= (A + B)x + (3A - B) \end{aligned}$$

$$A + B = 2$$

$$3A - B = 5$$

From $B = 2 - A$:

$$3A - (2 - A) = 5 \implies 4A - 2 = 5 \implies A = \frac{7}{4}$$

Then:

$$B = \frac{1}{4}$$

Final Result:

$$\frac{2x + 5}{(x - 1)(x + 3)} = \frac{7/4}{x - 1} + \frac{1/4}{x + 3}$$

5 2. Irreducible Quadratic Factors

Example:

$$\frac{x + 1}{(x^2 + 1)(x - 2)}$$

Solution:

$$\frac{Ax + B}{x^2 + 1} + \frac{C}{x - 2}$$

$$\begin{aligned} x + 1 &= (Ax + B)(x - 2) + C(x^2 + 1) \\ &= (A + C)x^2 + (-2A + B)x + (-2B + C) \end{aligned}$$

Setting up the system:

$$A + C = 0$$

$$-2A + B = 1$$

$$-2B + C = 1$$

Substituting gives:

$$A = -\frac{3}{5}, B = -\frac{1}{5}, C = \frac{3}{5}$$

Final Result:

$$\frac{x + 1}{(x^2 + 1)(x - 2)} = \frac{-\frac{3}{5}x - \frac{1}{5}}{x^2 + 1} + \frac{\frac{3}{5}}{x - 2}$$

6 3. Repeated Linear Factors

Example:

$$\frac{5}{(x-1)^2}$$

Solution:

$$\frac{A}{x-1} + \frac{B}{(x-1)^2}$$

Setting up the equation:

$$5 = A(x-1) + B$$

$$5 = Ax - A + B$$

Setting up the system:

$$A = 0, -A + B = 5$$

Results in:

$$A = 0, B = 5$$

Final Result:

$$\frac{5}{(x-1)^2} = \frac{0}{x-1} + \frac{5}{(x-1)^2}$$

7 4. Mixed Factors

Example:

$$\frac{3x+4}{(x-1)(x^2+1)}$$

Solution:

$$\frac{A}{x-1} + \frac{Bx+C}{x^2+1}$$

Setting up:

$$3x+4 = A(x^2+1) + (Bx+C)(x-1)$$

Combine terms:

$$3x+4 = (A+B)x^2 + (C-B)x + (A-C)$$

Setting up the system:

$$A+B=0$$

$$C-B=3$$

$$A-C=4$$

Results in:

$$A = \frac{7}{2}, B = -\frac{7}{2}, C = -\frac{1}{2}$$

Final Result:

$$\frac{3x+4}{(x-1)(x^2+1)} = \frac{\frac{7}{2}}{x-1} + \frac{-\frac{7}{2}x - \frac{1}{2}}{x^2+1}$$

8 Practice Questions

Write the following rational fractions in partial fraction decomposition:

1. $\frac{2x+1}{(x-1)(x+3)}$
2. $\frac{3x^2+2x+1}{(x^2+1)(x-2)}$
3. $\frac{5x}{(x+1)^2(x-2)}$
4. $\frac{x^2+4x+3}{(x-1)(x^2+1)}$
5. $\frac{7}{(x-2)(x^2+3)}$
6. $\frac{4x+1}{(x+2)(x-3)^2}$
7. $\frac{2x^2+3x+5}{(x-1)^2(x+4)}$
8. $\frac{x^3+x+1}{(x^2+2)(x+1)}$
9. $\frac{8}{(x-1)(x^2+4)}$
10. $\frac{6x+2}{(x+1)^2(x-2)}$

Difficult Mixed Partial Fraction Questions

1. Question 1:

$$\frac{5x+9}{(x^2+4)(x-3)}$$

Find the partial fraction decomposition.

2. Question 2:

$$\frac{4x^2+3x+2}{(x^2+1)(x-1)^2}$$

Find the partial fraction decomposition.

3. Question 3:

$$\frac{6x+8}{(x^2+1)(x+2)^2}$$

Find the partial fraction decomposition.

4. Question 4:

$$\frac{2x^2+5x+3}{(x-1)(x^2+1)(x+3)}$$

Find the partial fraction decomposition.

5. Question 5:

$$\frac{3x^3 + 2x^2 + 5}{(x^2 + 2)(x - 1)(x + 2)}$$

Find the partial fraction decomposition.